

The Standard Quantum Limit in Measurement Physics

A Scaffolded Treatise

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Abstract

The standard quantum limit (SQL) denotes several precision benchmarks whose form depends on the measurement model. In continuous position sensing, it arises from optimizing the competition between readout imprecision and quantum backaction for an uncorrelated meter. In phase estimation, it describes the $N^{-1/2}$ uncertainty scaling of independent probes, whereas in phase-preserving amplification it constrains the added noise required to preserve bosonic commutators. This article develops these limits using calibrated measurement records, spectral densities, linear response theory, and quantum estimation theory, then relates their underlying assumptions within a common measurement framework. Particular attention is given to the distinction among readout imprecision, added measurement noise, and total measured sensitivity. This distinction determines how a reported sub-SQL result should be interpreted, since its physical meaning depends on the chosen estimator, reference limit, bandwidth, spectral convention, and noise terms included in the comparison. The final sections show how probe preparation, engineered correlations, and the choice of observable alter the measurement assumptions from which each benchmark follows.

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1 Scope, terminology, and historical development

The following assumes familiarity with expectation values, variances, canonical commutators, and elementary Fourier transforms. Density matrices and Fisher information are introduced in later sections before they enter the analysis.

Precision measurement begins with a physical parameter (θ), such as displacement, force, strain, phase, frequency, or magnetic field, whose influence is encoded in a responsive physical system. Depending on the experiment, that response may appear as the motion of a mirror, a shift in a resonator, a change in an optical or microwave mode, or the collective evolution of an atomic or spin ensemble. A transduction mechanism then converts the physical response into an observable classical record, such as a voltage trace or a sequence of photon counts. Finally, a calibrated estimator interprets that record to produce an estimate ($\hat{\theta}$) together with its associated uncertainty.

The meaning of the term standard quantum limit is determined by the measurement chain, the resources available to it, and the adopted noise conventions. In a conventional continuous position measurement, the SQL arises from the balance between measurement imprecision and quantum backaction. Quantum metrology is the field that studies how quantum states, interactions, and measurement strategies determine the precision with which physical parameters can be estimated. Within this framework, an ensemble of N independent probes gives the statistical SQL, under which the estimation uncertainty decreases as $N^{-1/2}$. For a phase-preserving amplifier, a related quantum limit constrains the added noise required to preserve bosonic commutation relations. These limits may coexist within the same laboratory system, although a meaningful numerical comparison requires consistent definitions of the measured quantity, estimator, bandwidth, resource count, and spectral convention [1–5].

Table 1: Four uses of quantum limit language that require separate assumptions.

Setting	Bounded quantity	Assumptions that produce the standard benchmark	Common expression
Continuous position meter	Added displacement or force spectrum	Linear response, minimum uncertainty meter, and zero useful imprecision and backaction correlation	$\bar{S}_{xx}^{\text{SQL}} = \hbar \chi_m $
Statistical phase estimation	Estimator standard deviation	Independent probes, additive generator, and fixed single probe resource	$\Delta\theta \propto N^{-1/2}$
Phase preserving amplifier	Input referred added noise	Deterministic linear gain for both field quadratures	$n_{\text{add}} \geq 1/2$ at large gain
Correlation aware force sensing	Added force spectrum	Stationary linear sensing with optimized correlations and dissipative response	$\bar{S}_{FF}^{\text{add}} \geq \hbar \text{Im } \chi_m^{-1} $

The early theory of quantum limited measurement asked how a weak force could be inferred from a macroscopic oscillator after the measuring apparatus itself was treated as a quantum system. Braginskii showed that measurement disturbance belongs inside the sensitivity calculation [1]. Work with Vorontsov and Manukin then developed protocols that preserved selected observables across repeated measurements, which led to the theory of quantum nondemolition measurement [6, 7]. This approach became especially important for gravitational wave detection, where repeated interrogation must retain information about an extremely weak mechanical response [2, 8, 9].

Caves carried the same problem into optical interferometry by separating photon counting imprecision from radiation pressure backaction. Squeezed input altered their balance, and his amplifier analysis showed that commutator preservation imposes an added noise cost on phase preserving gain [10–12]. Later linear response and optomechanical treatments expressed these results in terms of detector susceptibility, spectral covariance, and exploitable correlations, which provides the framework used throughout this article [3, 13–17]. Spatial estimation extended the framework to imaging, where the parameter is encoded in the modal structure of the optical field [18].

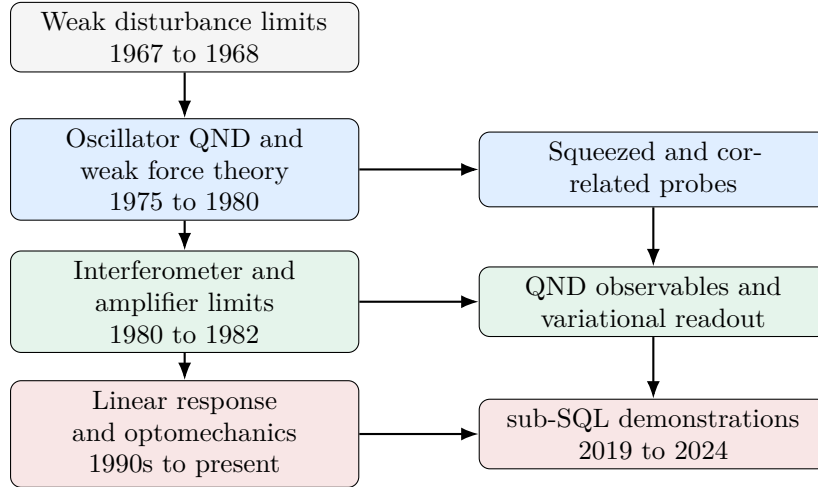


Figure 1: Conceptual lineage of the SQL. The arrows indicate dependence among ideas and carry no claim of exclusive authorship.

Table 2: Core vocabulary used throughout the article.

Term	Meaning
Imprecision	Noise referred to the input that obscures the measured variable in the readout record. Optical phase fluctuations and amplifier output noise are common sources.
Backaction	Fluctuating disturbance exerted by the meter on the measured system. Radiation pressure force and charge force noise are common examples.
Susceptibility	Complex linear response $\chi_m(\Omega)$ that maps applied force to displacement. Its magnitude sets signal gain, and its phase sets the correlation needed for cancellation.
Spectral density	Frequency resolved covariance of stationary fluctuations. Its numerical value depends on sidedness, symmetrization, and Fourier normalization.
Coherent state	Minimum uncertainty optical state whose two quadratures have equal vacuum variance. It supplies the common laser reference.
Squeezed state	State with reduced variance in one field quadrature and increased variance in its conjugate quadrature.
Quantum nondemolition measurement	Repeated measurement of an observable whose future values remain compatible with earlier measurements under the relevant dynamics and coupling.
Backaction evasion	Coupling or readout design that directs disturbance away from the signal observable or cancels its contribution in the estimator.
Quantum Fisher information	Largest local Fisher information available from a parameter dependent quantum state after optimization over quantum measurements.

Practical rule of thumb

Write the measured variable, resource, probe state, readout, bandwidth, and spectral convention beside every SQL formula. An unlabeled SQL curve carries incomplete physical information.

2 Measurement records, response functions, and spectra

The following assumes familiarity with complex Fourier amplitudes and the linear response of a driven harmonic oscillator. The symmetrized spectral density is introduced below, so no previous experience with quantum noise spectra is required.

2.1 From a physical disturbance to an estimator

A linear sensor can be understood as a chain of transduction steps. An external force perturbs a mechanical coordinate, whose response is acquired by an optical or electrical probe. The detector converts the probe response into a classical signal, and a filter processes that signal to estimate the applied force or another parameter of interest. Every stage contributes gain, bandwidth, loss, and noise. Expressing these contributions in terms of a common input variable allows their fluctuations and correlations to be combined within a single covariance model.

Let $x(t)$ denote the displacement of a mechanical mode and $F_{\text{sig}}(t)$ the force to be estimated. The Fourier domain response is

$$x(\Omega) = \chi_m(\Omega) [F_{\text{sig}}(\Omega) + F_{\text{ba}}(\Omega) + F_{\text{th}}(\Omega)], \quad (1)$$

where F_{ba} is meter backaction and F_{th} represents environmental force noise. For an oscillator with effective mass m , resonance Ω_m , and energy damping rate Γ_m ,

$$\chi_m(\Omega) = \frac{1}{m(\Omega_m^2 - \Omega^2 - i\Gamma_m\Omega)}. \quad (2)$$

The free mass approximation follows from $\Omega_m = 0$ and negligible damping over the signal band, which gives $\chi_m = -1/(m\Omega^2)$.

After calibration to displacement, the measured record is

$$y(\Omega) = x(\Omega) + x_{\text{imp}}(\Omega), \quad (3)$$

where x_{imp} is input referred imprecision. Equations (1) and (3) form the minimal model behind the continuous measurement SQL.

2.2 Symmetrized double-sided spectra

For stationary Hermitian fluctuations $\delta\hat{A}$ and $\delta\hat{B}$, this article uses

$$\bar{S}_{AB}(\Omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\Omega t} \left\langle \left\{ \delta\hat{A}(t), \delta\hat{B}(0) \right\} \right\rangle. \quad (4)$$

The bar denotes a symmetrized double-sided spectrum per unit angular frequency. For an auto-spectrum, or for an even classical symmetrized spectrum, the corresponding one-sided spectrum defined for $\Omega > 0$ is

$$S_{AB}^{(1)}(\Omega) = 2\bar{S}_{AB}(\Omega). \quad (5)$$

An amplitude spectral density is the square root of a power spectral density. Displacement amplitude spectra carry units of $\text{m}/\sqrt{\text{Hz}}$, force amplitude spectra carry $\text{N}/\sqrt{\text{Hz}}$, and strain amplitude spectra carry $1/\sqrt{\text{Hz}}$ after conversion to spectra per hertz.

Convention checkpoint

Every factor of two in an SQL spectrum depends on sidedness and symmetrization. Equations in this article use $\bar{S}$ for symmetrized double-sided spectra. A one sided result follows from Eq. (5).

At each angular frequency, the meter noise is characterized by the spectral density matrix

$$\bar{\mathbf{S}}_m(\Omega) = \begin{pmatrix} \bar{S}_{xx}^{\text{imp}} & \bar{S}_{xF} \\ \bar{S}_{xF}^* & \bar{S}_{FF}^{\text{ba}} \end{pmatrix}. \quad (6)$$

Referring the measurement record to an equivalent oscillator displacement yields the added displacement noise spectrum

$$\bar{S}_{xx}^{\text{add}}(\Omega) = \bar{S}_{xx}^{\text{imp}} + |\chi_m(\Omega)|^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re}[\chi_m^*(\Omega) \bar{S}_{xF}]. \quad (7)$$

After all noise contributions have been expressed in displacement units, the measured spectrum is

$$\bar{S}_{xx}^{\text{meas}} = \bar{S}_{xx}^{\text{intrinsic}} + \bar{S}_{xx}^{\text{add}} + \bar{S}_{xx}^{\text{tech}}, \quad (8)$$

where $\bar{S}_{xx}^{\text{intrinsic}}$ contains the equilibrium mechanical motion, including its thermal and zero-point contributions.

The cross spectrum $\bar{S}_{xF}$ between measurement imprecision and backaction controls the correlation contribution to the added noise. A phase quadrature readout of a coherent probe often provides little exploitable correlation. Variational homodyne detection can instead select a readout quadrature that exploits correlations associated with ponderomotive squeezing, allowing the correlation term in Eq. (7) to become negative over a selected frequency range and thereby reduce the added displacement noise.

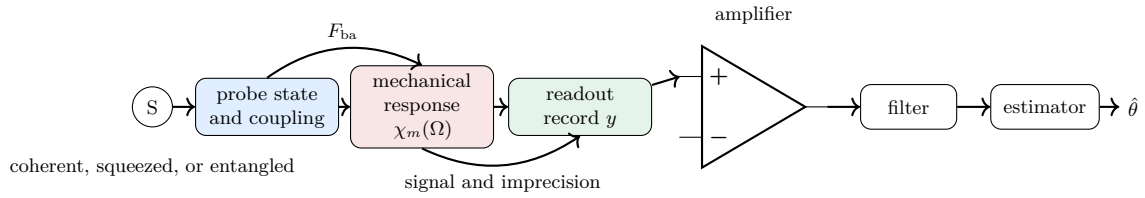


Figure 2: A calibrated quantum measurement chain. SQL performance depends on the full chain and on correlations retained by the estimator.

Practical rule of thumb

Refer every noise source to one variable before addition. Preserve the cross spectral terms until the chosen estimator and readout quadrature have been fixed.

3 Derivation of the continuous position and force SQL

Review Eqs. (2), (6), and (7). The derivation uses a minimum uncertainty linear meter with negligible reverse gain and states every spectral convention explicitly.

3.1 Quantum noise constraint

A general linear detector has output operator $\hat{I}$, backaction operator $\hat{F}$, forward susceptibility χ_{IF} , and reverse susceptibility χ_{FI} . Define

$$\tilde{\chi}_{IF}(\Omega) = \chi_{IF}(\Omega) - \chi_{FI}^*(\Omega). \quad (9)$$

The finite frequency quantum noise theorem can be written

$$\bar{S}_{II}\bar{S}_{FF} - |\bar{S}_{IF}|^2 \geq \left| \frac{\hbar\tilde{\chi}_{IF}}{2} \right|^2 \left[1 + \Delta\left(\frac{\bar{S}_{IF}}{\hbar\tilde{\chi}_{IF}/2} \right) \right], \quad (10)$$

where

$$\Delta[z] = \frac{|1 + z^2| - (1 + |z|^2)}{2}. \quad (11)$$

For the uncorrelated detector model used in the following SQL derivation, $\bar{S}_{IF} = 0$ and hence $\Delta[0] = 0$. The complete finite-frequency inequality remains necessary for detectors with general complex correlations, reverse gain, or nontrivial dynamical response [3].

For a canonical meter with negligible reverse gain, input referral by the forward gain and conditions that remove the correction reduce Eq. (10) to

$$\bar{S}_{xx}^{\text{imp}} \bar{S}_{FF}^{\text{ba}} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4}. \quad (12)$$

The interaction Hamiltonian and output response are

$$\hat{H}_{\text{int}} = -\hat{x}\hat{F}, \quad \hat{I}(\Omega) = \hat{I}_0(\Omega) + \chi_{IF}(\Omega)\hat{x}(\Omega). \quad (13)$$

Equation (12) supplies the transparent algebra for the standard uncorrelated SQL. The full theorem is required for a general detector or a realizability proof [19, 20].

3.2 Optimization for an uncorrelated meter

Set $\bar{S}_{xF} = 0$ and parameterize measurement strength by a positive dimensionless variable λ ,

$$\bar{S}_{xx}^{\text{imp}} = \frac{A}{\lambda}, \quad \bar{S}_{FF}^{\text{ba}} = B\lambda. \quad (14)$$

Stronger coupling carries more information and lowers imprecision, and the same coupling raises the fluctuating force. The added displacement spectrum is

$$\bar{S}_{xx}^{\text{add}} = \frac{A}{\lambda} + |\chi_m|^2 B\lambda. \quad (15)$$

Differentiation with respect to λ gives

$$\lambda_{\text{opt}} = \sqrt{\frac{A}{|\chi_m|^2 B}}, \quad (16)$$

and

$$\bar{S}_{xx,\text{min}}^{\text{add}} = 2|\chi_m|\sqrt{AB}. \quad (17)$$

For a meter that saturates Eq. (12), $AB = \hbar^2/4$, which yields

$$\boxed{\bar{S}_{xx}^{\text{SQL}}(\Omega) = \hbar|\chi_m(\Omega)|}. \quad (18)$$

The force referred form follows from division by $|\chi_m|^2$,

$$\boxed{\bar{S}_{FF}^{\text{SQL}}(\Omega) = \frac{\hbar}{|\chi_m(\Omega)|}}. \quad (19)$$

At the optimum of this uncorrelated model, the imprecision and backaction contributions are equal; other architectures can reach their optima with unequal contributions.

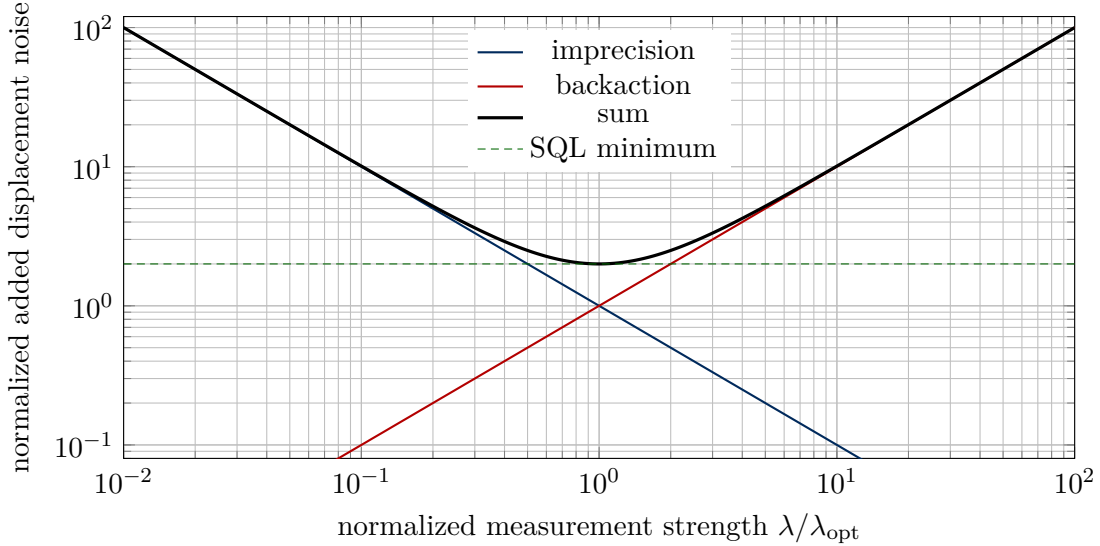


Figure 3: Normalized imprecision and backaction tradeoff for an uncorrelated linear meter.

3.3 Free mass and resonant oscillator

For a free mass, $|\chi_m| = 1/(m\Omega^2)$, so

$$\bar{S}_{xx}^{\text{SQL}} = \frac{\hbar}{m\Omega^2}, \quad \bar{S}_{FF}^{\text{SQL}} = \hbar m\Omega^2. \quad (20)$$

For a lightly damped oscillator at resonance,

$$|\chi_m(\Omega_m)| = \frac{1}{m\Gamma_m\Omega_m}, \quad \bar{S}_{FF}^{\text{SQL}}(\Omega_m) = \hbar m\Gamma_m\Omega_m. \quad (21)$$

One sided spectra are twice these values. The square root amplitude spectra therefore acquire a factor $\sqrt{2}$.

The equilibrium one sided thermal force spectrum of a viscously damped oscillator can be written

$$S_{FF}^{\text{th},(1)}(\Omega) = 2m\Gamma_m\hbar\Omega \coth\left(\frac{\hbar\Omega}{2k_B T}\right). \quad (22)$$

Its classical limit is $4m\Gamma_mk_BT$. Quantum measurement noise becomes experimentally visible after thermal, laser, electronic, and calibration contributions approach the relevant SQL spectrum.

3.4 Frequency dependent optimum

The oscillator susceptibility changes rapidly near resonance, so a fixed probe strength reaches the optimum over a limited band. Define

$$\tilde{\chi}(r) = \frac{1}{\sqrt{(1-r^2)^2 + (\gamma r)^2}}, \quad r = \frac{\Omega}{\Omega_m}, \quad (23)$$

and normalized added noise

$$N(r, \lambda) = \lambda^{-1} + \lambda|\tilde{\chi}(r)|^2. \quad (24)$$

With $u = \lambda|\tilde{\chi}|$, the SQL ratio is

$$\frac{N}{N_{\text{SQL}}} = \frac{1}{2} (u + u^{-1}) \geq 1. \quad (25)$$

Equality gives $\lambda_{\text{opt}} = 1/|\tilde{\chi}|$. Figure 4 evaluates this expression directly with PGFPlots and requires no external data file.

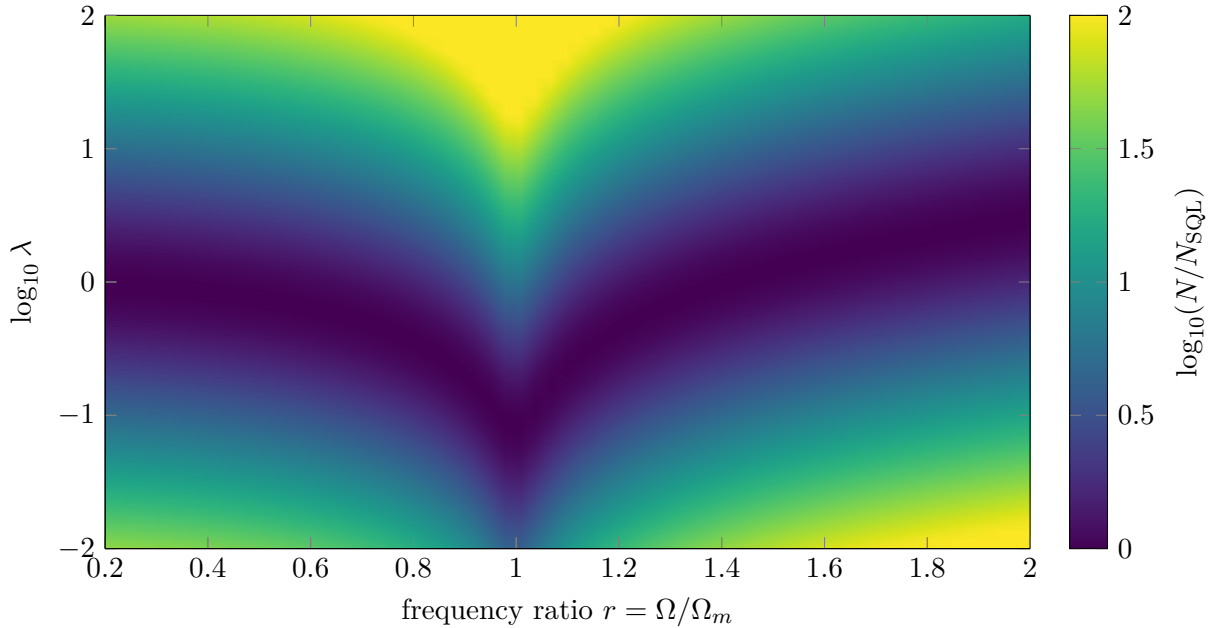


Figure 4: Excess added noise above the uncorrelated SQL for damping ratio $\gamma = 0.08$. The valley follows $\lambda_{\text{opt}} = 1/|\tilde{\chi}|$, and the plotted surface is generated analytically from the provided mathematical formulas to ensure computational reproducibility.

3.5 SQL and the correlation aware lower bound

Useful $\bar{S}_{xF}$ can lower Eq. (7) below Eq. (18). Stationary linear force sensing remains constrained by dissipation. In the same double-sided convention, common forms of the ultimate added noise bounds are

$$\bar{S}_{xx}^{\text{add}} \geq \hbar |\text{Im } \chi_m|, \quad \bar{S}_{FF}^{\text{add}} \geq \hbar |\text{Im } \chi_m^{-1}|. \quad (26)$$

For viscous damping, the force bound reduces to the dissipative quantum limit (DQL),

$$\bar{S}_{FF}^{\text{DQL}}(\Omega) = \hbar m \Gamma_m |\Omega|. \quad (27)$$

At mechanical resonance χ_m is nearly imaginary, so the uncorrelated displacement SQL and the dissipative displacement bound coincide. Away from resonance, their separation quantifies the improvement available from correlations. An ideal lossless free mass has a real susceptibility, and Eq. (26) then signals the importance of finite observation time, finite probe energy, optical loss, and other resource constraints that lie outside the stationary dissipative bound [21–23].

Practical rule of thumb

Balance imprecision and backaction to locate the uncorrelated SQL. Use the full covariance matrix and the dissipative response to evaluate any claimed advantage from correlations.

4 Optical interferometers and cavity optomechanics

This section assumes familiarity with optical phase, coherent state photon statistics, and the susceptibility of a harmonic oscillator. The spectral relations needed for the discussion were developed in Section 2.

4.1 Coherent light in a Michelson position meter

A differential arm displacement x changes the round trip phase by

$$\phi = 2k_0 x = \frac{4\pi x}{\lambda_0}. \quad (28)$$

For N detected photons in a coherent state, phase imprecision scales as

$$\Delta\phi_{\text{imp}} \sim N^{-1/2}, \quad \Delta x_{\text{imp}} \sim \frac{\lambda_0}{4\pi\sqrt{N}}. \quad (29)$$

With optical power P , integration time τ , and carrier frequency ω_0 , the detected photon number scales as $N \sim P\tau/(\hbar\omega_0)$. The displacement imprecision spectrum therefore scales as P^{-1} .

Coherent state amplitude quadrature fluctuations produce photon flux fluctuations around a bright carrier. Their fluctuating momentum transfer produces a radiation pressure force spectrum proportional to P . The phase quadrature produces readout imprecision. The equivalent displacement spectrum from force noise is multiplied by $|\chi_m|^2$, so a compact model is

$$\bar{S}_{xx}^{\text{add}}(\Omega; P) = \frac{\alpha(\Omega)}{P} + \beta(\Omega)P|\chi_m(\Omega)|^2. \quad (30)$$

The optical commutator fixes $\alpha\beta$ for an ideal coherent probe. Minimization over P reproduces Eq. (18). Strain calibration in a simple differential arm model uses $x = hL$, which gives $S_{hh} = S_{xx}/L^2$ after the effective differential mechanical mode has been defined.

The full interferometer response is shaped by optical storage in the arm and signal recycling cavities, together with the selected homodyne quadrature and the losses accumulated before detection. These elements rotate and attenuate the output noise ellipse in a frequency dependent manner. A filter cavity applies a corresponding rotation to the injected squeezed state, aligning its reduced noise quadrature with phase readout at high frequency and with radiation pressure suppression at low frequency [24–26].

4.2 Cavity optomechanical implementation

For a single optical and mechanical mode, the dispersive Hamiltonian is

$$\hat{H} = \hbar\omega_c\hat{a}^\dagger\hat{a} + \hbar\Omega_m\hat{b}^\dagger\hat{b} - \hbar g_0\hat{a}^\dagger\hat{a}(\hat{b} + \hat{b}^\dagger). \quad (31)$$

The single photon coupling g_0 gives the cavity frequency shift produced by zero point motion. Driving the cavity to mean photon number n_c produces the linearized coupling $G = g_0\sqrt{n_c}$. For a resonant drive in the unresolved sideband measurement limit, a representative measurement rate is

$$\Gamma_{\text{meas}} \simeq \frac{4\eta G^2}{\kappa}, \quad (32)$$

where η is total detection efficiency and κ is the cavity energy decay rate. Larger n_c increases information rate and radiation pressure shot noise together. The cooperativity

$$C = \frac{4G^2}{\kappa\Gamma_m} \quad (33)$$

compares optomechanical measurement dynamics with mechanical damping. Quantum cooperativity also accounts for thermal occupation and indicates whether quantum backaction can compete with thermalization [16, 27].

The oscillator zero point displacement is

$$x_{\text{zpf}} = \sqrt{\frac{\hbar}{2m\Omega_m}}. \quad (34)$$

For $k_B T \gg \hbar\Omega_m$, the thermal occupation is $n_{\text{th}} \simeq k_B T / (\hbar\Omega_m)$. The product $\Gamma_m n_{\text{th}}$ estimates a thermal heating rate. At fixed quality factor $Q = \Omega_m / \Gamma_m$,

$$\Gamma_m n_{\text{th}} \simeq \frac{k_B T}{\hbar Q}. \quad (35)$$

Frequency alone therefore gives no reduction of this rate at fixed Q . Material loss, temperature, surface contamination, optical absorption, and mode shape determine the realized thermal environment.

Experiments have observed radiation pressure backaction, measured forces close to the SQL, and used optomechanical correlations to obtain total off resonant force and displacement sensitivity below the uncorrelated SQL [21, 28, 29]. The last statement is stronger than a report of imprecision below an SQL scale. It includes the calibrated intrinsic mechanical motion, measurement imprecision, backaction, and their correlations in the measured band.

Practical rule of thumb

Increasing optical power improves phase readout until radiation pressure, absorption, heating, or dynamical backaction erases the gain. Track detected efficiency, intracavity power, and mechanical thermalization as separate control variables.

5 From continuous records to Fisher information

This section assumes a basic understanding of density matrices ρ , including their role as positive, unit trace operators that describe pure and mixed quantum states. Familiarity with conditional probability and variance will also be useful. The estimators and Fisher information quantities needed later are introduced as they arise.

5.1 Classical estimators and local information

A measurement with outcomes k and parameter dependent probabilities $p(k|\theta)$ has classical Fisher information

$$F_C(\theta) = \sum_k p(k|\theta) \left[\frac{\partial}{\partial \theta} \ln p(k|\theta) \right]^2. \quad (36)$$

For ν independent repetitions, a locally unbiased estimator obeys the Cramér–Rao inequality

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\nu F_C(\theta)}. \quad (37)$$

The bound describes local precision near an operating point where the estimator is unbiased and the likelihood changes smoothly with θ . Finite data can introduce estimator bias, nuisance parameters can couple the desired parameter to unknown quantities, and periodic variables can create equivalent points in parameter space. These cases require finite sample, multiparameter, or Bayesian extensions matched to the actual inference problem.

A quantum state ρ_θ and a positive operator valued measure $\{E_k\}$ give

$$p(k|\theta) = \text{Tr}(\rho_\theta E_k), \quad E_k \succeq 0, \quad \sum_k E_k = \mathbb{I}. \quad (38)$$

Optimization over all measurements yields quantum Fisher information. The symmetric logarithmic derivative L_θ satisfies

$$\partial_\theta \rho_\theta = \frac{1}{2} (L_\theta \rho_\theta + \rho_\theta L_\theta), \quad (39)$$

and

$$F_Q(\theta) = \text{Tr}(\rho_\theta L_\theta^2), \quad F_C(\theta) \leq F_Q(\theta). \quad (40)$$

The quantum Cramér–Rao bound is

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\nu F_Q(\theta)}. \quad (41)$$

Attainability depends on the state family, available measurement, local operating point, and multiparameter compatibility [4, 5, 30].

For unitary encoding generated by $\hat{G}$ and a mixed state $\rho = \sum_j p_j |j\rangle\langle j|$,

$$F_Q[\rho, \hat{G}] = 2 \sum_{j,k: p_j + p_k > 0} \frac{(p_j - p_k)^2}{p_j + p_k} |\langle j|\hat{G}|k\rangle|^2. \quad (42)$$

For a pure state $|\psi\rangle$, this expression reduces to

$$F_Q = 4(\Delta \hat{G})_\psi^2. \quad (43)$$

Quantum Fisher information grows when state preparation and interaction engineering increase the rate at which the state changes with the parameter. For unitary encoding, squeezing and entanglement can increase the variance of the generator. The final measurement then determines how much of that available information reaches the classical record and the estimator.

5.2 Independent probes and the statistical SQL

Let the parameter be encoded by

$$\hat{U}_\theta = e^{-i\theta\hat{G}}, \quad \hat{G} = \sum_{j=1}^N \hat{g}_j. \quad (44)$$

For N independent probes in a product state, local generator variances add,

$$(\Delta\hat{G})^2 = N(\Delta\hat{g})^2, \quad F_Q \propto N, \quad \Delta\theta \propto N^{-1/2}. \quad (45)$$

If the j th generator has spectral range $\Delta g_j = g_{j,\max} - g_{j,\min}$, every separable state obeys

$$F_Q[\rho_{\text{sep}}, \hat{G}] \leq \sum_{j=1}^N (\Delta g_j)^2. \quad (46)$$

Identical qubits with $\hat{g}_j = \sigma_z^{(j)}/2$ satisfy $F_Q \leq N$, so

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\nu N}. \quad (47)$$

This inequality gives a precise restricted state QCRB and supports the phrase statistical SQL [31, 32].

An ideal GHZ state places amplitude at the extremal eigenvalues of the total generator and reaches

$$F_Q = N^2, \quad \text{Var}(\hat{\theta}) \geq \frac{1}{\nu N^2}. \quad (48)$$

This Heisenberg scaling assumes a consistent resource count and coherent local encoding. Sequential coherent reuse can reproduce the same scaling under a resource definition that counts every pass. Independent loss and Markovian dephasing commonly return the asymptotic scaling toward $N^{-1/2}$, leaving a useful constant factor advantage over separable probes [33–35].

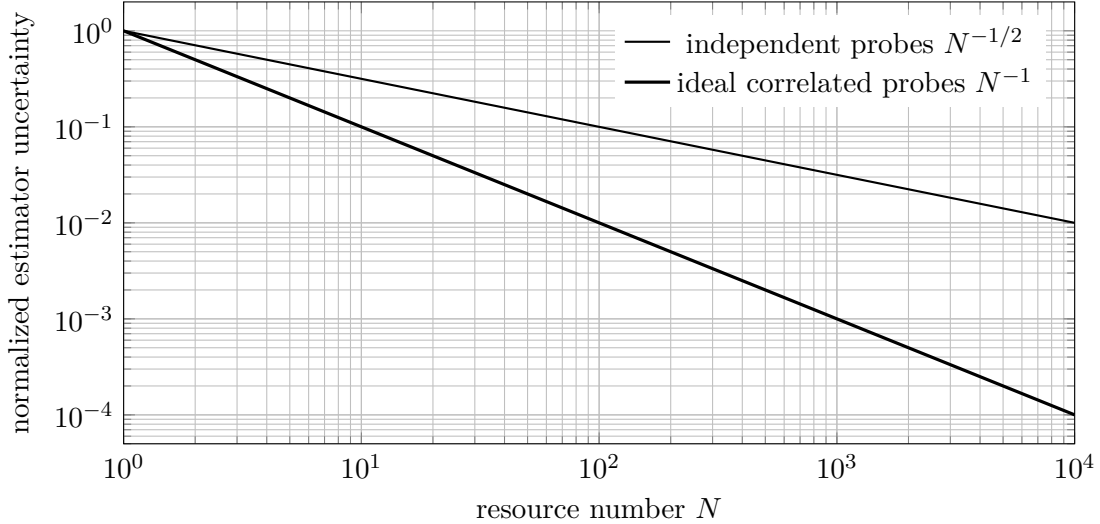


Figure 5: Ideal resource scaling. Loss, decoherence, and preparation overhead determine the realized advantage.

5.3 The bridge from spectra to estimators

The preceding continuous measurement theory describes a noisy record indexed by time or Fourier frequency. Fisher information describes the distinguishability of records or states indexed by a parameter. A stationary Gaussian record connects the two descriptions directly.

Consider

$$y(\Omega) = \theta s(\Omega) + n(\Omega), \quad (49)$$

where s is a known finite duration signal template and n is zero mean Gaussian noise with double-sided spectrum $\bar{S}_{nn}$. The matched filter estimator is

$$\hat{\theta} = \frac{\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{s^*(\Omega)y(\Omega)}{\bar{S}_{nn}(\Omega)}}{\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}}. \quad (50)$$

Its variance is

$$\text{Var}(\hat{\theta}) = \left[\int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)} \right]^{-1}, \quad (51)$$

which is the inverse Fisher information for this Gaussian model. Force estimation uses $s(\Omega) = \chi_m(\Omega)F_0(\Omega)$ after displacement calibration. Strain estimation uses the interferometer response to the chosen waveform.

Probe strength changes $\bar{S}_{nn}$ by changing imprecision, backaction, and their cross spectrum. Minimizing a narrowband noise spectrum therefore maximizes the Fisher information for a signal in that band. Broadband optimization weights every frequency by $|s|^2/\bar{S}_{nn}$, so the smallest pointwise spectrum and the smallest waveform estimation error can select different instrument settings. Functional quantum bounds extend this logic to time dependent forces and show how quantum noise cancellation and smoothing can approach the waveform QCRB [36]. The statistical and continuous SQLs meet at this estimator layer.

Practical rule of thumb

Use $F_Q \propto N$ as the precise separable probe benchmark. Use matched filter Fisher information for a continuous waveform. Keep resource counting, bandwidth, and estimator bias fixed during every comparison.

6 Physical routes below the standard benchmark

Review the meter covariance matrix in Eq. (6), the added noise expression in Eq. (7), and the distinction between the SQL and Eq. (26).

6.1 Squeezed quadratures

Define optical quadratures by

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \quad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}, \quad [\hat{X}_1, \hat{X}_2] = \frac{i}{2}. \quad (52)$$

Vacuum and coherent states have variance $1/4$ in each quadrature. A squeezed Gaussian state has covariance

$$\mathbf{V}(r, \phi) = \frac{1}{4} \mathbf{R}(\phi) \begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix} \mathbf{R}^\top(\phi), \quad (53)$$

where r is squeeze strength and ϕ is squeeze angle. A fixed phase squeezed input lowers high frequency readout imprecision and raises the conjugate amplitude fluctuations that drive low frequency radiation pressure. Frequency dependent rotation $\phi(\Omega)$ aligns the reduced variance with the frequency dependent signal quadrature [11, 24, 25].

Loss replaces a fraction of the state with vacuum,

$$\mathbf{V}_{\text{obs}} = \eta \mathbf{V}(r, \phi) + (1 - \eta) \frac{\mathbf{I}}{4}. \quad (54)$$

Large antisqueezing also converts small phase jitter into observed noise. Propagation loss, mode mismatch, photodetection efficiency, filter cavity loss, and squeeze angle control therefore set the realized improvement.

6.2 Correlation engineering and variational readout

Choose the phase of $\bar{S}_{xF}$ so that

$$\text{Re}[\chi_m^* \bar{S}_{xF}] < 0. \quad (55)$$

The cross term in Eq. (7) then removes part of the backaction contribution from the record. Completing the square gives

$$\bar{S}_{xx}^{\text{add}} = \left(\bar{S}_{xx}^{\text{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\text{ba}}} \right) + \bar{S}_{FF}^{\text{ba}} \left| \chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\text{ba}}} \right|^2. \quad (56)$$

Within the reduced detector model, Eq. (12) gives

$$\bar{S}_{xx}^{\text{add}} \geq \frac{\hbar^2}{4\bar{S}_{FF}^{\text{ba}}} + \bar{S}_{FF}^{\text{ba}} \left| \chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\text{ba}}} \right|^2. \quad (57)$$

Useful anticorrelation lowers the second term in Eq. (57), although the desired cross spectrum must still be produced by a causal detector with finite gain and stable dynamics. Reverse gain and dynamical backaction modify the mechanical response, optical loss weakens the correlation before detection, and the accessible homodyne angles restrict its usable phase. These conditions determine whether the algebraic cancellation can persist across a useful bandwidth.

Variational readout implements this idea by selecting a frequency dependent homodyne quadrature. Ponderomotive interaction creates correlations between optical amplitude and phase, and the readout selects the combination that retains signal with reduced backaction contamination. Broadband displacement experiments have used these correlations to move below the ordinary uncorrelated sensitivity curve [21, 37].

Coherent quantum noise cancellation introduces an auxiliary response with an effective sign opposite to the sensor response. Proper matching cancels backaction at the level of coherent dynamics, which is closely related to negative mass reference frames and quantum mechanics free subsystems [38].

6.3 Quantum nondemolition observables and backaction evasion

An ideal QND observable $\hat{A}$ satisfies

$$[\hat{A}(t_i), \hat{A}(t_j)] = 0 \quad (58)$$

at the sampling times, and the measurement interaction preserves that observable. A sufficient ideal coupling condition is

$$[\hat{A}, \hat{H}_{\text{int}}] = 0. \quad (59)$$

The disturbance enters a conjugate variable whose uncertainty is absent from the later target measurement [8, 9, 14].

For a harmonic oscillator, define slowly varying rotating quadratures $\hat{X}$ and $\hat{P}$ with $[\hat{X}, \hat{P}] = i$. A balanced two tone interaction isolates one rotating quadrature within the rotating wave approximation, so the meter disturbance enters its conjugate partner. Counterrotating terms couple that disturbance back into the measured quadrature, damping mixes the quadratures during the observation interval, and pump imbalance opens a direct backaction channel. Sideband resolution therefore sets how closely the experiment approaches the ideal QND interaction [39–41].

For a free mass, its later position depends on the momentum it carried earlier. Any momentum disturbance introduced by the first measurement therefore appears in subsequent position readouts as backaction noise. A speed meter senses a velocity like observable, and variational readout estimates a combination of output quadratures whose backaction contribution is reduced. Contractive state analyses and realizable measurement models also show that the free mass SQL depends on state preparation and measurement protocol [42, 43]. Equation (20) should therefore be identified as the SQL of the stated uncorrelated continuous position meter.

6.4 Phase preserving amplification

A deterministic phase preserving bosonic amplifier with power gain G has the canonical map

$$\hat{a}_{\text{out}} = \sqrt{G} \hat{a}_{\text{in}} + \sqrt{G-1} \hat{b}_{\text{in}}^\dagger. \quad (60)$$

The auxiliary mode preserves $[\hat{a}_{\text{out}}, \hat{a}_{\text{out}}^\dagger] = 1$. With the auxiliary input in vacuum, the minimum input referred added noise is

$$n_{\text{add}} \geq \frac{1}{2} \left(1 - \frac{1}{G} \right), \quad (61)$$

which approaches one half quantum at large gain. Phase sensitive amplification can add zero noise to one selected quadrature as it deamplifies the conjugate quadrature. It gives up simultaneous phase preserving gain for both quadratures [12, 44].

Table 3: Representative strategies for surpassing a standard benchmark.

Method	Change to the measurement model	Representative platform
Phase squeezing	Reduces detected phase quadrature variance	Optical phase sensors and interferometers
Frequency dependent squeezing	Rotates the noise ellipse across Fourier frequency	Gravitational wave detectors
Variational readout	Selects a correlated output quadrature that cancels backaction in the estimator	Interferometers and optomechanics

Method	Change to the measurement model	Representative platform
Backaction evasion	Couples the meter to one rotating mechanical quadrature	Microwave and optical optomechanics
Speed meter	Measures a velocity like observable with repeated interaction geometry	Gravitational wave detector concepts
Coherent noise cancellation	Matches an auxiliary response of opposite sign to the sensor response	Optomechanical force sensors
Spin squeezing	Reduces collective spin variance at fixed mean spin signal	Atomic clocks and magnetometers
Entangled probes	Increases total generator variance above separable scaling	Optical and atomic phase estimation
QND measurement	Selects an observable compatible with its future values	Oscillators, photons, spins, and qubits
Phase sensitive amplification	Amplifies one chosen field quadrature	Microwave qubit readout and radio frequency sensing

Practical rule of thumb

Identify the assumption changed by each sub-SQL technique. Squeezing changes the probe covariance, variational readout changes the measured quadrature, QND changes the observable, and entanglement changes generator variance.

7 Applications and evidence standards

Review the difference among imprecision, added noise, and total measured noise. A claim about one level gives limited information about the others.

7.1 Gravitational wave interferometry

Laser interferometers infer strain from differential arm motion over kilometer baselines. Optical phase noise dominates much of the high frequency quantum noise, and radiation pressure motion gains importance at lower frequencies. Squeezed vacuum injection first improved detector sensitivity beyond the coherent shot noise level, Advanced LIGO later operated with squeezed vacuum, and frequency dependent squeezing then provided broadband quantum noise reduction [26, 45–47].

The LIGO Livingston analysis reported the quantum noise contribution below the free mass SQL from 35 to 75 Hz with a maximum improvement of 3 dB near 50 Hz after installation of the A+ frequency dependent squeezing system [48, 49]. Classical noise exceeded quantum noise in that low frequency band, so the reported quantum spectrum was inferred by calibrated model subtraction and dedicated squeeze angle measurements. The distinction preserves the importance of the result and states its experimental scope accurately.

7.2 Optomechanical and atomic force sensing

These experiments mark three distinct stages in the development of quantum limited force sensing. Murch and collaborators directly observed radiation pressure backaction, establishing the meter force as a measurable physical contribution [28]. Schreppler and collaborators then resolved the tradeoff between imprecision and backaction in a calibrated force measurement [29]. Mason and collaborators used the resulting correlation constructively, reducing the total off resonance force and displacement noise below the uncorrelated SQL after thermal motion and detector noise had been included [21].

In atomic force microscopy, the force estimate is inferred from cantilever motion, so cantilever thermal fluctuations and optical readout noise enter the same calibrated record. Increasing optical power can reduce imprecision, yet it can also increase radiation pressure and local heating. The tip sample interaction adds another force channel that is absent from a free cantilever calibration [50,51]. Pooser and collaborators used truncated nonlinear interferometry to reduce the optical readout noise below the coherent reference and reached $1.7 \text{ fm}/\sqrt{\text{Hz}}$ [52]. That result establishes quantum enhanced optical readout. A total force SQL test also requires calibrated thermal motion, backaction, and tip sample force noise over the same measurement band.

Magnetic resonance force microscopy reached single electron spin detection by coupling spin dynamics to a sensitive cantilever [53]. That milestone concerns signal detection and force sensitivity. Its relation to an SQL requires the complete quantum and thermal noise budget for the chosen force estimator.

7.3 Spin ensembles and clocks

For N uncorrelated two level atoms polarized along one collective spin direction, transverse projection noise scales as $\sqrt{N}$ and phase uncertainty scales as $N^{-1/2}$. The Wineland metrological squeezing parameter is

$$\xi_R^2 = \frac{N(\Delta J_\perp)^2}{|\langle \mathbf{J} \rangle|^2}. \quad (62)$$

Values $\xi_R^2 < 1$ indicate phase sensitivity below the coherent spin state SQL after contrast loss is included. The parameter connects reduced collective variance with a usable signal slope and remains central to atomic clocks, magnetometers, and atom interferometers [5, 54, 55].

7.4 Spatial parameter estimation and quantum imaging

In quantum imaging, the unknown quantity may be a source position, source separation, object displacement, spatial phase, or another parameter encoded in the optical field. Diffraction determines which spatial modes reach the detector, and estimation theory determines how accurately the parameter can be inferred from those modes. For a single spatial coordinate r , the quantum Cramér–Rao bound gives

$$\text{Var}(\hat{r}) \geq \frac{1}{\nu F_Q(r)}. \quad (63)$$

For several spatial parameters, the corresponding quantum Fisher information matrix determines the covariance bounds and reveals any incompatibility among their optimal measurements.

Spatial mode measurements retain sensitivity to source separation because the parameter changes the modal composition of the field even after the intensity profile varies only weakly. A squeezed or number correlated probe can add a second advantage by reducing fluctuations in the modes that carry the parameter derivative. Loss and detector inefficiency mix those informative modes with vacuum and

reduce the gain, so the benchmark must use the same photon budget and receiver measurement as the experiment [56].

In active imaging, nonclassical spatial modes can reduce uncertainty in beam displacement and optical phase estimates by concentrating correlations in the modes that carry the parameter response. The attainable gain depends on how well propagation and detection preserve those correlations, so comparisons should hold photon number, state-preparation cost, and receiver measurement fixed. The resulting advantage may originate from the probe state, the modal measurement, or both, and should be evaluated against matched classical and quantum estimation bounds.

Practical rule of thumb

When evaluating a published, proposed, or experimentally reported superresolution result, the analysis should state the estimated parameter, optical state, measurement strategy, photon budget, and reference bound. A comparison with the Rayleigh criterion evaluates resolution under a specified imaging procedure, whereas comparison with the quantum Cramér–Rao bound evaluates the estimator precision permitted by the probe state, propagation channel, and available measurement.

7.5 A hierarchy of experimental claims

Table 4: Evidence required for increasingly strong SQL statements.

Claim	Required comparison	Common missing element
Sub shot noise readout	Detected readout variance against a coherent or vacuum reference at equal signal resource	Loss corrected resource accounting
Imprecision below SQL scale	Calibrated input referred imprecision against a stated SQL displacement scale	Backaction and thermal noise
Quantum noise below SQL	Calibrated sum of quantum imprecision, backaction, and correlations against the same SQL convention	Classical noise and model uncertainty
Total sensitivity below SQL	Measured total estimator noise, including thermal and technical contributions, against a calibrated SQL curve	Independent calibration and uncertainty propagation
Metrological gain below separable bound	Estimator variance or Fisher information against equal resource separable probes	State preparation cost, contrast, and bias

Table 5: How the SQL enters representative platforms.

Platform	Measured parameter	Standard balance or resource restriction	References
Laser interferometer	Differential displacement or strain	Phase imprecision against radiation pressure backaction	[11, 26, 48]

Platform	Measured parameter	Standard balance or resource restriction	References
Cavity optomechanics	Displacement or force	Optical imprecision against photon number backaction	[16, 21]
Atomic force microscope	Cantilever motion and force	Optical readout, cantilever disturbance, and thermal force	[50, 52]
Spin ensemble	Phase, frequency, or field	Independent projection noise against collective correlations	[54, 55]
Microwave amplifier	Field quadratures	Phase preserving gain against commutator preserving added noise	[12, 44]
Quantum phase estimation	Optical or atomic phase	Separable generator variance against correlated generator variance	[30, 33]
Quantum imaging	Source position, separation, or another spatial parameter	Direct imaging compared with quantum-optimal spatial-mode measurements; nonclassical-probe bounds require a separately specified active-imaging model	[18, 56]

Practical rule of thumb

Reserve the phrase total sensitivity below the SQL for a comparison that includes intrinsic mechanical motion, technical noise, measurement imprecision, backaction, correlations, and calibration uncertainties in the same units and bandwidth.

8 Experimental design and numerical verification

Before beginning the numerical analysis, specify the parameter to be estimated, the estimator bandwidth, and the spectral convention. These choices define the physical problem and provide the basis for a meaningful optimization.

8.1 Noise budget and calibration sequence

A defensible SQL comparison can be built with the following sequence.

1. Define the input variable and estimator. State whether the result refers to displacement, force, strain, phase, frequency, or field.
2. Measure or model the complex transfer function from the input variable to the recorded channel. Preserve its phase.
3. Fix the Fourier normalization, spectral sidedness, symmetrization, resolution bandwidth, windowing, and averaging method.
4. Calibrate imprecision with the physical coupling suppressed or independently varied.

5. Calibrate backaction by varying probe strength and measuring the driven response or induced heating.
6. Estimate the cross spectrum with phase information. A magnitude squared coherence by itself discards the sign needed for cancellation.
7. Measure thermal and technical backgrounds and propagate their calibration uncertainty.
8. Construct the SQL from the measured effective susceptibility and the same convention used for data.
9. Repeat the comparison across probe strengths and readout quadratures to show the expected covariance geometry.

For a total measured displacement spectrum,

$$\bar{S}_{xx}^{\text{meas}} = \bar{S}_{xx}^{\text{intrinsic}} + \bar{S}_{xx}^{\text{tech}} + \bar{S}_{xx}^{\text{add}}. \quad (64)$$

The intrinsic spectrum can be decomposed into thermal and zero-point contributions when the adopted noise model treats them as separate terms. In some analyses, modeled classical backgrounds are subtracted from the measured spectrum to infer the remaining quantum-noise contribution, as occurs in low-frequency interferometer studies. Any conclusion that this inferred quantum component reaches or falls below a stated SQL should specify the background-subtraction procedure, the assumptions entering the noise model, and the uncertainty propagated into the inferred spectrum.

8.2 Control parameters for an SQL heat map

The analytic heat map in Fig. 4 has clear controls. The damping ratio γ sets resonance width, the plotted frequency interval sets the mechanical response range, the range of $\log_{10} \lambda$ sets the available measurement strength, and the sample count sets rendering density. These values appear as control knobs in the preamble. A laboratory version can replace Eq. (23) with measured complex susceptibility and replace Eq. (24) with the measured covariance matrix.

Define a dimensionless excess added noise metric

$$H(\Omega, \lambda) = \log_{10} \left[\frac{\bar{S}_{xx}^{\text{add}}(\Omega, \lambda)}{\bar{S}_{xx}^{\text{SQL}}(\Omega)} \right]. \quad (65)$$

$H = 0$ lies on the selected SQL, $H > 0$ lies above it, and $H < 0$ lies below it. A plot with correlations must evaluate the complex cross term in Eq. (7). Using $|\bar{S}_{xF}|$ without its phase can produce a physically incorrect valley.

8.3 Common diagnostic failures

Several checks catch common errors before a sub-SQL conclusion is drawn.

- Dimensional consistency must hold after input referral. A force spectrum multiplied by $|\chi_m|^2$ has displacement spectrum units.
- The optimum of Eq. (15) must occur where its two positive terms are equal.
- Conversion from a double-sided to a one sided power spectrum multiplies by two, and the corresponding amplitude spectrum multiplies by $\sqrt{2}$.
- A squeezed variance quoted in decibels uses $10 \log_{10}(V/V_0)$. An amplitude ratio uses $20 \log_{10}(A/A_0)$.

- A sub-SQL quantum component can coexist with total detector noise above the SQL.
- A narrowband noise minimum can differ from the setting that maximizes broadband waveform Fisher information.

Practical rule of thumb

Treat susceptibility phase, cross spectral phase, sidedness, and calibration covariance as primary data. These quantities decide whether a plotted sub-SQL valley represents physical sensitivity or a convention mismatch.

9 Synthesis

SQL takes on a definite physical meaning once the measured quantity, available resources, probe state, estimator, bandwidth, and noise convention are specified. For an uncorrelated continuous position measurement, the symmetrized double-sided displacement spectrum reaches $\hbar|\chi_m|$. An ensemble of independent probes follows the familiar $N^{-1/2}$ scaling of statistical uncertainty, whereas a phase-preserving amplifier approaches one-half quantum of input-referred added noise at large gain. Force sensing that exploits imprecision–backaction correlations is subject to a lower dissipative bound set by the imaginary part of the inverse mechanical susceptibility. Although these results share a common foundation in quantum mechanics, each applies to a different measurement task, so a direct comparison requires consistent experimental conditions, definitions, and conventions.

Quantum-enhanced sensing improves precision by changing how quantum uncertainty enters the measurement. Squeezing concentrates fluctuations in the variable that matters most for the estimate, and variational readout uses correlations between imprecision and backaction so that part of the disturbance can be removed from the inferred signal. Backaction-evading schemes and speed meters go further by changing which mechanical quantity is monitored, thereby preventing the measurement disturbance from returning in the same form. Entanglement enlarges the response of the collective probe to the parameter of interest, and QND interactions preserve the measured observable so that it can be interrogated repeatedly. These methods all respect the quantum uncertainty relations, yet they exceed reference limits that assume uncorrelated probes, fixed observables, or simpler measurement couplings.

A convincing demonstration consists of defining the reference curve and the estimator compared with it. The resource count and spectral convention set the numerical scale of the benchmark, and the calibrated noise model determines which fluctuations enter the reported sensitivity. Propagating calibration uncertainty through the same estimator then permits comparisons across otherwise different sensing platforms without conflating readout imprecision with total sensitivity.

Practical rule of thumb

SQL is a calibrated design benchmark. A complete sub-SQL result identifies the assumption that has changed and demonstrates the improvement using the full estimator noise budget.

A Fourier and spectral conversion reference

With

$$A(\Omega) = \int_{-\infty}^{\infty} A(t)e^{i\Omega t} dt, \quad A(t) = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} A(\Omega)e^{-i\Omega t}, \quad (66)$$

Eq. (4) gives the double-sided symmetrized spectrum. For a real stationary process,

$$\langle A^2(t) \rangle = \int_{-\infty}^{\infty} \frac{d\Omega}{2\pi} \bar{S}_{AA}(\Omega) = \int_0^{\infty} \frac{d\Omega}{2\pi} S_{AA}^{(1)}(\Omega). \quad (67)$$

Spectra per hertz use $f = \Omega/(2\pi)$ and satisfy $S^{[f]}(f) df = S^{[\Omega]}(\Omega) d\Omega$. Stating the integration measure prevents a hidden factor of 2π .

B Analytic heat map definition

The heat map uses $r = \Omega/\Omega_m$, $\ell = \log_{10} \lambda$, and damping ratio γ . Its plotted function is

$$H(r, \ell) = \log_{10} \left[\frac{10^{-\ell} + 10^{\ell} |\tilde{\chi}(r)|^2}{2|\tilde{\chi}(r)|} \right]. \quad (68)$$

The arithmetic geometric mean inequality gives $H \geq 0$. Equality occurs at

$$\ell_{\text{opt}}(r) = -\log_{10} |\tilde{\chi}(r)|. \quad (69)$$

This analytic construction replaces the external `sql_heatmap.dat` dependency and compiles from the supplied TeX source and bibliography database.

C Compact formula sheet

Table 6: Frequently used formulas and their scope.

Formula	Scope
$\bar{S}_{xx}^{\text{add}} = \bar{S}_{xx}^{\text{imp}} + \chi_m ^2 \bar{S}_{FF}^{\text{ba}} + 2 \text{Re}(\chi_m^* \bar{S}_{xF})$	Linear input referred displacement measurement
$\bar{S}_{xx}^{\text{SQL}} = \hbar \chi_m $	Ideal uncorrelated meter, symmetrized double-sided spectrum
$\bar{S}_{FF}^{\text{SQL}} = \hbar / \chi_m $	Force referred form of the same SQL
$\bar{S}_{FF}^{\text{add}} \geq \hbar \text{Im} \chi_m^{-1} $	Correlation aware stationary dissipative force bound
$F_Q = 4(\Delta G)^2$	Pure state under unitary parameter encoding
$F_Q \leq \sum_j (\Delta g_j)^2$	Separable probes with bounded local generator ranges
$F_Q = N^2$	Ideal N qubit GHZ probe under the stated local generator
$n_{\text{add}} \geq (1 - G^{-1})/2$	Deterministic phase preserving bosonic amplifier
$\xi_R^2 = N(\Delta J_{\perp})^2 / \langle \mathbf{J} \rangle ^2$	Metrological spin squeezing parameter

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